

Geometry and Calculus for Computing

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Module 1

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Angles, triangles and trigonometry

Lesson 1.0 Introduction

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Course structure and navigation

Reading • 15 min

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How to learn effectively on this course

Reading • 15 min

Introduction to the course

Video • 1 min

Course Syllabus

Reading • 10 min

Lesson 1.1 Introduction to Angles, triangles and trigonometry

Introduction to triangles and trigonometry

Video • 15 min

The circle

Video • 8 min

From the circle to the sine and cosine graphs

Video • 9 min

Introducing the tangent

Video • 5 min

Applying trigonometry to solve triangle problems

Dialogue • 30 min

Introduction to triangles and trigonometry

Practice Assignment • 30 min

Lesson 1.2 Generic Triangles

Applications of sine and cosine rules – examples

Video • 10 min

Sine and cosine rules

Practice Assignment • 30 min

Lesson 1.3 Trigonometric Functions

Unit circumference and definition of trigonometric functions for every angle

Video • 9 min

Plotting tan

Video • 2 min

Unit circumference and definition of trigonometric functions for every angle

Practice Assignment • 30 min

Trigonometric functions, plots and properties

Video • 19 min

Trigonometric functions, plots and properties

Practice Assignment • 30 min

Lesson 1.4 Summary and Assessments

Check your understanding: End of module 1

Graded Assignment • 20 min

Summary

Reading • 10 min

Module 2

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Graph sketching and kinematics

Module 3

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Exponential and logarithmic functions

Module 4

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Limits and differentiation

How to learn effectively on this course

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This short guide is here to support you as you go through the course, especially if you’re studying on your own. You'll find some simple, research-informed strategies to help you stay motivated and get more out of your study time, without feeling overwhelmed.

Staying motivated

Many students say they need motivation to learn, but **motivation is unpredictable** – it comes and goes. What really keeps you moving forward is building habits. Creating small, manageable learning routines helps you to stay on track even when you're not feeling particularly inspired. Start by keeping your study sessions short and focused, just enough to get into the flow without feeling overwhelmed. You can try using the **pomodoro technique** (25 minutes of focused work followed by a 5-minute break) to stay on track. Celebrate small wins along the way – progress is more powerful than perfection. Even completing one item is a success and helps build momentum.

To make your learning habit stick, set a fixed study schedule and pair it with an existing routine, like watching a video with your morning coffee or reviewing notes before bed. Linking a new habit to something you already do makes it easier to remember and follow through – this is called **habit stacking**. In addition, use the platform's tools to track your progress, for example, click 'Mark as complete' for each activity so you can easily see how far you've come. Staying accountable also helps. Connect with others in discussion forums or ask a friend or a family member to check in on your progress. And remember, motivation grows when learning feels meaningful – try connecting what you’re learning to real-world topics that matter to you.

Motivation is not something you wait for. It's something you create. Take one minute now to write down **why** you joined this course and **when and where** you plan to study each day. Keep it somewhere visible – you'll thank yourself later.

Learning to learn

Learning doesn’t happen just by reading or watching. It happens when you actively work with the material. One of the most effective ways to do this is by testing yourself soon after encountering something new. Even if you get it wrong, that moment of effort strengthens the learning process by creating connections in your brain.

In fact, learners who test themselves, whether with closed or open-ended questions or by explaining ideas in their own words, retain far more than those who simply review. This approach is especially powerful because of something called the **forgetting curve**, which shows that we begin to forget information almost immediately after learning it. When you test yourself, you interrupt that forgetting curve and help your brain hold on to what matters. Coming back to the material again later strengthens learning even more, this means revisiting what you’ve learned after a break, instead of trying to memorise it all at once.

You can also boost understanding by using real-life examples to make abstract ideas more relatable, or by mixing different topics or types of questions within the same session. These strategies might feel more effortful, but that's exactly why they work. The active mental effort you put in while learning, such as testing yourself after encountering new concepts, helps generate focused attention. That attention signals to your brain, 'This is important–keep it'. Without that signal, your brain is more likely to let new information fade away. Passive review alone isn't enough, it's the effortful engagement that helps learning stick.

If you're interested in exploring the concepts behind these strategies, such as, **retrieval practice, spaced practice, elaboration, metacognition, interleaving and concrete examples**, you’re more than welcome to. However, that’s not the intention of this course. You don't need to memorise these terms; what's most important is understanding that **how** you study is just as important as **what** you study.

Your study journal

We commend you to start your own study journal, in a format of your choosing, which could be traditional pen and paper or an electronic version. Your journal will enable you to collate information, monitor progress and organise your notes. You can use it later for reference when completing graded assignments.

The journal is for your reference only and will not be assessed.

Which format should you choose?

The format that you are most comfortable with!

You may prefer to use pen and paper, but it is worth bearing in mind that electronic documents are easily searchable and can store links.

Online documents are preferable to static documents as you will be able to access them from any computer – handy if you study in more than one place! Options include a live Word doc, Google doc, OneNote, a personal blog, Evernote or any other document of your choice.

Examples

Journal entries could include the following:

- Links to websites and articles
- Your own thoughts and reflections
- Your responses to directed activities
- Notes on readings and videos.

To give you an idea of how you could set yours out, here is [a screencast example of a study journal](#) .

Next steps

Start your own study journal using one of the tools suggested.

Now post an entry in your study journal answering the following questions:

- How will you record reflective practice in your journal?
- Which journal tool will you use?

Let’s get started on the course.

Go to next item

✓ Completed